

Chapter 1 — Practice Quiz

IP Static Routing & Troubleshooting | SRWE Modules 15-16 | No Answers

A. Static Route Types & Concepts

1.
What are the 4 types of static routes by purpose?
2.
What are the 3 ways to specify the next hop when configuring a static route?
3.
On what type of link should a directly connected static route (exit-interface only) be used?
4.
A fully specified static route must be used in two scenarios. Name them.
5.
What command must be enabled on a router before IPv6 static routes will work?

B. Command Syntax

6.
Write the full syntax for an IPv4 static route command.
7.
Write the full syntax for an IPv6 static route command.
8.
IPv4 default static route via next-hop 10.0.0.1.
9.
IPv6 default static route via next-hop 2001:db8::1.
10.
IPv4 static host route to 209.165.200.238 via 198.51.100.2.
11.
IPv4 floating default static route via backup 10.10.10.2 with AD 5.

C. Administrative Distance

12.
What does Administrative Distance (AD) represent?
13.
Default ADs: connected route? static route? OSPF? RIP?
14.
To make a floating route, higher or lower AD than the primary?
15.
Does a floating route appear in the table while the primary is up? Why?

D. Default Routes

16.
What is an IPv4 default static route informally called?
17.
What network address and subnet mask are used for an IPv4 default route?
18.
What prefix/length is used for an IPv6 default route?
- 19.

In show ip route, how do you identify the candidate default route?

20.

Give two typical scenarios where a default static route is used.

E. Summary & Host Routes

21.

What two conditions must be met to summarize multiple static routes?

22.

Summarize 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24 into one route.

23.

What subnet mask identifies an IPv4 host route?

24.

What prefix length identifies an IPv6 host route?

25.

Name the three ways a host route can appear in a routing table.

F. Packet Forwarding

26.

When a packet arrives at a router, what are the 3 possible outcomes of the routing table lookup?

27.

Does the source/destination IP address change hop to hop?

28.

What changes at each hop as a packet travels?

29.

On the final router, what protocol finds the destination MAC address?

30.

What does a router do if no ARP entry exists for the destination?

G. Troubleshooting

31.

List 5 common IOS troubleshooting commands and what each checks.

32.

PC1 cannot reach PC3. Pings from R1 to R2 and R1 to R3 succeed. Where is the problem?

33.

What is the recommended troubleshooting strategy across multiple routers?

34.

Give four common reasons a network fails (routing-related).

35.

What does show cdp neighbors tell you?

H. Scenario / Application

36.

Three routes to 192.168.1.0/24, 2.0/24, 3.0/24 all via 172.16.2.2 — can they be summarized?

37.

Primary vs backup: interpret 'ip route 0.0.0.0 0.0.0.0 203.0.113.1' and '... 198.51.100.1 10'.

38.

For an IPv6 static route with link-local next-hop, why must the exit interface be specified?

39.

Write an IPv6 fully specified route to 2001:db8:acad:1::/64 via S0/1/0, next-hop fe80::1.

40.

S 10.1.1.0/24 [1/0] via 192.168.99.1 — 192.168.99.1 is not connected. What is wrong?

Quick Fire (True / False)

41.

A static route is learned automatically by the router. T / F

42.

IPv6 static routes use the same syntax as IPv4 with ipv6 instead of ip. T / F

43.

The default AD for a static route is 0. T / F

44.

A directly connected static route is recommended for Ethernet interfaces. T / F

45.

A default route matches only packets with destination 0.0.0.0. T / F

46.

A floating static route is used as the primary path. T / F

47.

The show ip route static command displays only the static routes. T / F

48.

Route summarization reduces the size of the routing table. T / F

49.

A host route uses a /24 mask for IPv4. T / F

50.

With no matching route and no default, the router forwards to all interfaces. T / F